

Chapitre 7 : Composantes d'interface utilisateur

Solutions des exercices

Exercice 1

```
public class Temperature {
    double valeur;

    public Temperature() {
        valeur = 0;
    }

    double c2f(){
        return (valeur * 9 / 5 + 32);
    }

    double f2c(){
        return ((valeur - 32) * 5 / 9);
    }
}

private class BtnConvertirActionListener implements ActionListener {
    public void actionPerformed(ActionEvent arg0) {

        Temperature maTemp = new Temperature();

        if(comboBox.getSelectedIndex()== 0){
            maTemp.valeur = Double.parseDouble(textField.getText());
            label.setText(""+ maTemp.c2f());
        }
        else {
            maTemp.valeur = Double.parseDouble(textField.getText());
            label.setText(""+ maTemp.f2c());
        }
    }
}

private class ComboBoxActionListener implements ActionListener {
    public void actionPerformed(ActionEvent e) {

        if(comboBox.getSelectedIndex()== 0){
            lblInput.setText("Température en Celsius");
            lblOutput.setText("Temp\u00E9rature en Fahrenheit");
        }
        else {
            lblInput.setText("Temp\u00E9rature en Fahrenheit");
            lblOutput.setText("Température en Celsius");
        }
    }
}
```

```
    }  
}  
}
```

Exercise 2

```
public class Appel {  
  
    int categorie;  
    int duree;  
  
    public Appel() {  
    }  
  
    public double cout(){  
  
        if (categorie == 0) {  
            return duree * 0.01;  
        }  
        else if (categorie == 1) {  
            return duree * 0.08;  
        }  
        else if (categorie == 2) {  
            return duree * 0.20;  
        }  
        else if (categorie == 3) {  
            return duree * 0.25;  
        }  
        else if (categorie == 4) {  
            return duree * 0.40;  
        }  
        else if (categorie == 5) {  
            return duree * 0.70;  
        }  
        else {  
            return 0;  
        }  
  
    }  
}
```

```

}

private class BtnCalculerActionListener implements ActionListener {
    public void actionPerformed(ActionEvent arg0) {
        Appel unAppel = new Appel();

        unAppel.duree = Integer.parseInt(txtDuree.getText());
        unAppel.categorie = comboBox.getSelectedIndex();

        lblCout.setText("" + unAppel.cout());

    }
}

```

Exercise 3

```

private class BtnCalculerActionListener implements ActionListener {
    public void actionPerformed(ActionEvent arg0) {

        Appel unAppel = new Appel();

        unAppel.duree = Integer.parseInt(txtDuree.getText());

        if (rdbtnLocal.isSelected()) {
            unAppel.categorie = 0;
        }
        else if (rdbtnCanada.isSelected()) {
            unAppel.categorie = 1;
        }
        else if (rdbtnAmerique.isSelected()) {
            unAppel.categorie = 2;
        }
        else if (rdbtnEurope.isSelected()) {
            unAppel.categorie = 3;
        }
        else if (rdbtnAsie.isSelected()) {
            unAppel.categorie = 4;
        }
        else if (rdbtnAfrique.isSelected()) {
            unAppel.categorie = 5;
        }
        else {

```

```

        unAppel.categorie = 0;
    }

    lblCout.setText("" + unAppel.cout());
}
}
}

```

Exercise 4

```

public class Consommation {

    double litre;
    double km;

    double consommation() {
        return km/litre;
    }
}

private class BtnCalculerActionListener implements ActionListener {
    public void actionPerformed(ActionEvent arg0) {
        Consommation monAuto = new Consommation();
        monAuto.km = Double.parseDouble(textField.getText());
        monAuto.litre = Double.parseDouble(textField_1.getText());
        textArea.setText("Kilomètres parcourus : " + monAuto.km + " km\nLitres de carburant : " + monAuto.litre +
" l\nConsommation : " + monAuto.consommodation()+" km/l");
    }
}

```